

Project Title : Strategic product placement analysis

(unveiling sales impact with tableau visualization)

Team ID : LTVIP2026TMIDS55906

Team Leader : Chenna Priyanka

Team Member : G Sailaja

Team Member : Esambadi Pavan Kalyan

Team Member : G Kalyani

1.INTRODUCTION :

1.1 Project Overview :

The project “Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization” focuses on analyzing how the positioning of products within a retail store affects customer purchasing behavior and overall sales performance. Using interactive Tableau dashboards, the project visualizes sales trends, product visibility patterns, and placement effectiveness across different store zones.

By transforming raw sales data into insightful visual representations, the project enables retailers to understand which product placements drive higher engagement, better conversions, and improved revenue. This data-driven approach helps optimize store layouts and enhances strategic decision-making.

1.2 Purpose of the Project:

The primary purpose of this project is to **evaluate and enhance the effectiveness of product placement strategies** through data visualization and analytics. The project aims to:

- Identify how shelf position, aisle location, and promotional zones influence sales.
- Provide retailers with actionable insights to optimize product layouts.
- Improve customer buying experience by placing high-demand products more strategically.
- Support data-driven decision-making for maximizing sales, reducing operational inefficiencies, and improving store performance.

Overall, the project serves as a decision-support tool that empowers businesses to refine their merchandising strategies and achieve measurable sales improvements.

2.IDEATION PHASE :

2.1 Problem Statement :

Retailers often lack clear insights into how product placement inside a store influences customer purchases and overall sales. Traditional sales reports show numbers but do not reveal how shelf position, aisle location, or promotional zones affect product performance. This gap leads to ineffective store layouts and missed opportunities to boost sales.

The project aims to solve this problem by using **Tableau visualizations** to analyze sales data and identify patterns related to product placement. The insights help retailers make better merchandising decisions and improve product visibility to maximize sales impact.

Problem Statement (PS)	I am (Customer)	I’m trying to	But	Because	Which makes me feel

PS-1	A retail store manager.	Understand which product placements increase sales.	I can't identify which shelf or zone performs best.	Sales data is not clearly visualized.	Confused and unable to make effective placement decisions.
PS-2	A retail business owner.	Improve product visibility and sales.	I don't know which locations impact customer buying.	There is no dashboard showing placement-based sales insights.	Frustrated and limited in planning store layout.

2.2 Empathy map Canvas :

The Empathy Map Canvas captures the needs and expectations of retail managers and data analysts who seek better clarity on how product placement impacts sales. They believe that visual dashboards can simplify complex sales data and provide meaningful insights into customer buying behavior. Currently, they rely on traditional sales reports and manual analysis, which often fail to reveal placement-related patterns.

This lack of clarity creates frustration and uncertainty when making layout or merchandising decisions. The canvas highlights the need for simple, visual, and actionable insights that help stakeholders optimize product placement, enhance customer engagement, and ultimately improve overall sales performance.

Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization



2.3 Brainstorming :

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Our team collaborated virtually and reviewed the WHC Sites 2019 dataset. After thorough discussion, we selected the problem statement:

"To analyze and visualize the global distribution, area, and danger status of UNESCO World Heritage Sites using Tableau for insightful storytelling."

Step-2: Brainstorm, Idea Listing and Grouping

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

Step-1: Team Gathering, Collaboration and Select the Problem Statement :

Problem Statement Selected:

Retail and e-commerce businesses often struggle to understand how the placement of products within stores or on digital storefronts influences sales performance. Without proper analytics, decisions around shelf placement, category grouping, and promotional positioning remain guess-based rather than data-driven.

Objective of the Project:

To analyze product placement strategies and identify which shelf positions, zones, or display layouts lead to maximum sales. The project aims to deliver a Tableau-based interactive dashboard to visualize patterns, trends, and high-performing product positions.

Key Discussion Points During Team Collaboration:

- Many retailers fail to measure placement impact despite collecting sales data.
- Understanding placement effectiveness can boost revenue significantly.
- Visualization using Tableau can make insights more interactive and decision-friendly.
- Project can create real-world value for retail domains, supermarkets, and online platforms.



Step-2: Brainstorm, Idea Listing and Grouping :

Raw Ideas Generated:

1. Analyze sales trends based on shelf levels (top, eye-level, bottom).
2. Compare sales between end-cap displays vs normal shelves.
3. Identify customer buying patterns influenced by product adjacency.
4. Visualize heatmaps showing high-selling store zones.
5. Determine correlation between product category placement and sales lift.
6. Integrate promotional placement impact (discount vs location).
7. Suggest optimal product arrangement for maximum visibility.
8. Build an interactive Tableau dashboard for real-time visualization.
9. Use KPIs like conversion rate, sales uplift, customer footfall patterns.
10. Recommend model for ideal store layout using insights.

Grouped Ideas

Group A – Sales & Placement Analysis

- Shelf-level sales evaluation
- Zone-based heatmaps

- End-cap vs regular shelf comparison
- Product adjacency impact analysis

Group B – Customer Behavior Insights

- Buying trend visualization
- Category-wise placement effectiveness
- Seasonal impact on placement

Group C – Visualization & Decision Tools

- Tableau dashboards
- Key metrics and KPIs
- Recommended layout strategies
- Interactive filters for store managers

Step-3: Idea Prioritization :

Prioritization Criteria Used:

- Feasibility
- Data availability
- Value creation for retailers
- Ease of visualization in Tableau
- Impact on decision-making

Final Prioritized Solution:

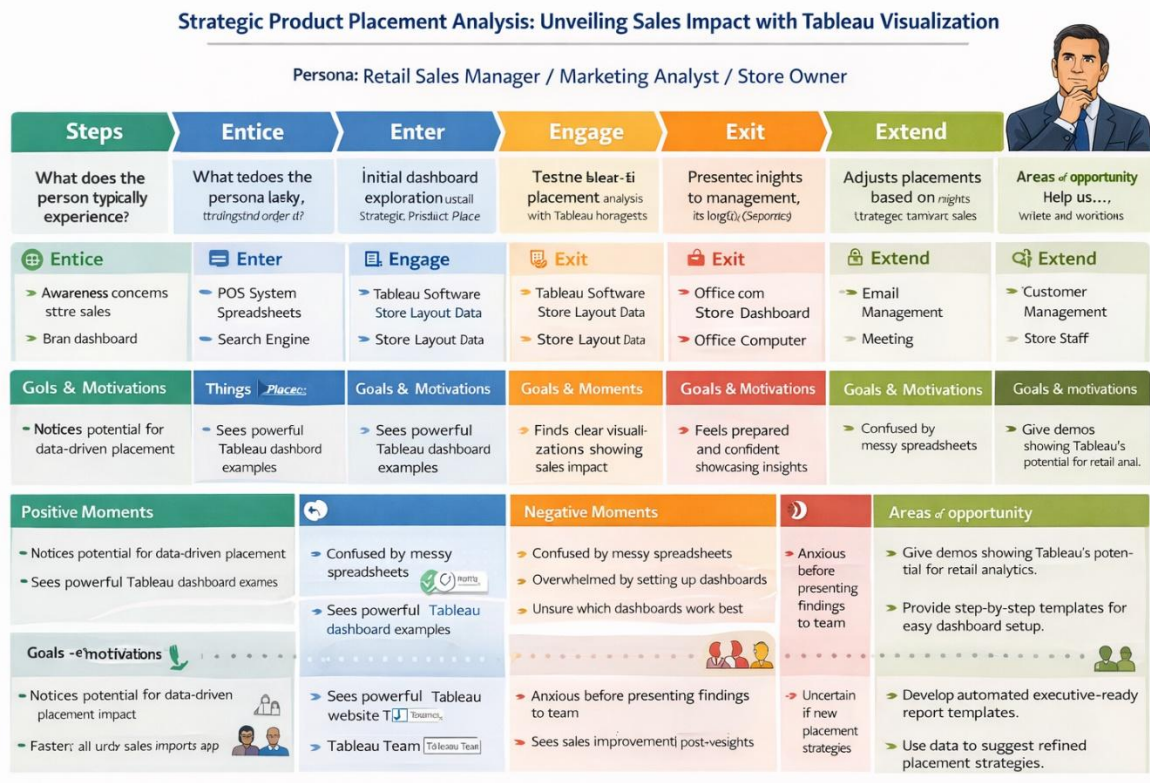
The team will focus on creating a comprehensive Tableau dashboard that reveals the impact of product placement on sales using:

- Shelf-level performance charts
- Zone heatmaps
- End-cap effectiveness analysis
- Category & adjacency insights
- KPIs for quick decision-making

The goal is to deliver an actionable visualization system enabling businesses to optimize product placement and maximize sales.

3.REQUIREMENT ANALYSIS :

3.1 Customer Journey Map :



3.2 Solution Requirement :

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import & Integration	Import sales data (CSV/Excel) Connect to POS system data Import store layout / planogram data
FR-2	Data Processing & Cleaning	Handle missing values, duplicates Standardize product/category fields Map product IDs to shelf positions

FR-3	Dashboard Visualization	Sales performance visualization charts Product placement heatmaps Before/After placement comparison Interactive filters (time, product, zone)
FR-4	Insights & Reporting	Automated insights on top-performing zones Generate summary reports Export dashboard insights as PDF
FR-5	User Access & Sharing	Publish dashboard on Tableau Public Share access links with stakeholders Role-based viewing permissions

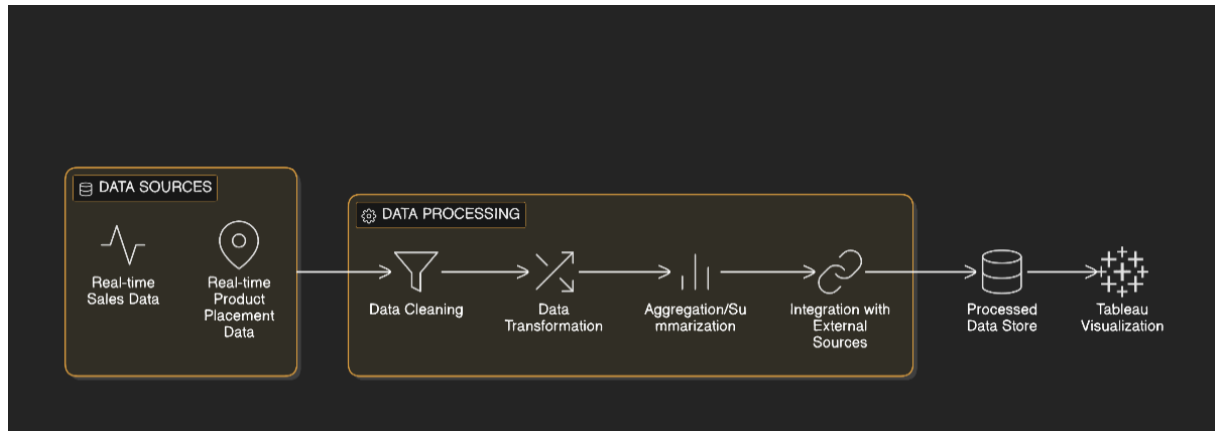
Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard should be simple, intuitive, and easy for retail managers to understand.
NFR-2	Reliability	System must load data and visuals accurately without errors or corruption.
NFR-3	Performance	Dashboard must load within 3–5 seconds even with large datasets.
NFR-4	Scalability	Supports increasing data size and additional store locations without performance loss.

3.3 Data Flow Diagram :

The Data Flow Diagram (DFD) for the project “**Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization**” represents how sales data moves through the system to produce meaningful insights.



3.4 Technology Stack :

The project “**Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization**” uses a streamlined technology stack to collect, process, analyze, and visualize sales data for better product placement decisions.

1. Data Source

Excel / CSV Files – Used to store sales data, product details, and placement information.

2. Data Processing

Python (Pandas, NumPy) – Cleans, preprocesses, and prepares the datasets for analysis.

Jupyter Notebook / VS Code – Environment used to run data analysis scripts.

3. Data Visualization

Tableau Public / Tableau Desktop – Core visualization tool used to build dashboards, heatmaps, KPIs, and sales impact visualizations.

4. Supporting Tools

MS Excel – Initial data formatting and quick calculations.

GitHub / Cloud Storage – Storing project files, documentation, and datasets.

5. Deployment / Sharing

Tableau Public Link / Web Embed – Used to share the final interactive dashboard with stakeholders.

4. PROJECT DESIGN :

4.1 Problem Solution Fit :

Retail businesses often struggle to understand how **product placement inside the store impacts sales**, because sales reports are scattered, layout data is not visual, and insights are not clear.

This project provides a **Tableau-based visualization system** that combines sales data with store layout to reveal **which product placements drive the highest sales**, enabling data-driven decisions to optimize store performance.

Purpose:

1. Identify the store zones and shelf positions that increase sales.
2. Visualize sales and placement data through interactive Tableau dashboards.
3. Compare product performance before and after placement changes.
4. Enable data-driven decision-making for retail managers and analysts.

Template:

Problem-Solution fit canvas 2.0			Purpose / Vision		
Define CS, fit into CC	1. CUSTOMER SEGMENT(S) Who is your customer? - Retail store owners - Retail sales managers - Marketing analysts - Merchandising teams - Data-driven decision-makers in retail	CS	Explore AS, differentiate	6. CUSTOMER CONSTRAINTS What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices. - Understand which product placements increase sales. - Identify high-performing store zones and low-performing ones. - Visualize before/after impact of placement changes. - Improve product positioning strategies to boost revenue. - Present insights clearly to management.	CC
				5. AVAILABLE SOLUTIONS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking - Basic POS reports with limited insights - Excel-based manual analysis - Traditional static reports - Third-party analytics tools (expensive or not specific to product placement)	AS
Focus on J&P, tap into BE, understand RC	2. JOBS-TO-BE-DONE / PROBLEMS Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides. - Understand which product placements increase sales - Identify high-performing store zones and low-performing ones - Visualize before/after impact of placement changes - Improve product positioning strategies to boost revenue - Present insights clearly to management	J&P	Focus on J&P, tap into BE, understand RC	9. PROBLEM ROOT CAUSE - Sales data and layout data are not connected or visualized together - Manual analysis hides patterns and trends - No tool to show real-time placement effectiveness - Lack of interactive visual analytics	RC
				7. BEHAVIOUR What does your customer do to address the problem and get the job done? i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greengear) - Retail managers rely heavily on experience rather than data - Frequent guesswork when changing product placements - Reviewing raw spreadsheets instead of visual dashboards	BE
Identify strong TR & EM	3. TRIGGERS What triggers customers to act? i.e. seeing their neighbour installing - Sudden drop or inconsistency in product sales - Management asking for proof of placement decisions	TR	Extract online & offline CH of BE	10. YOUR SOLUTION - An interactive Tableau dashboard integrating sales and store layout data - Visual heatmaps showing high and low performing store zones - Before-and-after comparison for product placement changes	SL
	4. EMOTIONS: BEFORE / AFTER Before Feels confused due to unclear data and scattered information	EM		8. CHANNELS of BEHAVIOUR 8.1 ONLINE - Tableau Public dashboard access - Email communication for sharing insights 8.2 OFFLINE - In-store meetings with managers or merchandisers - Printed reports or presentations during discussions	CH

References:

1. <https://www.ideahackers.network/problem-solution-fit-canvas/>
<https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>

4.2 Proposed Solution :

The proposed solution for “**Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization**” is a data-driven analytical system that identifies how product

placement influences sales performance. The solution uses structured data analysis and interactive visual dashboards to help retailers make smarter decisions.

S.No.	Parameter	Description
	Problem Statement (Problem to be solved)	Retailers struggle to understand how product placement affects sales due to scattered and non-visual data.
	Idea / Solution description	A Tableau-based interactive dashboard that visualizes the impact of product placement on sales performance.
	Novelty / Uniqueness	It uniquely integrates store layout and sales data to reveal real-time placement effectiveness through visual analytics.
	Social Impact / Customer Satisfaction	Enhances shopping experience and sales efficiency by ensuring products are placed where customers find them easily.
	Business Model (Revenue Model)	Revenue can be generated through subscription-based analytics dashboards and customized retail insights services
	Scalability of the Solution	The system can easily scale across multiple stores, larger datasets, and additional product categories.

4.3 Solution Architecture :

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Integrate sales and store layout data into a unified analytical system.
- Use Tableau dashboards to visually analyze product placement effectiveness.
- Provide interactive filters and comparison views to evaluate before–after placement impact.
- Generate automated insights and reports to support data-driven decision-making.

Example - Solution Architecture Diagram:

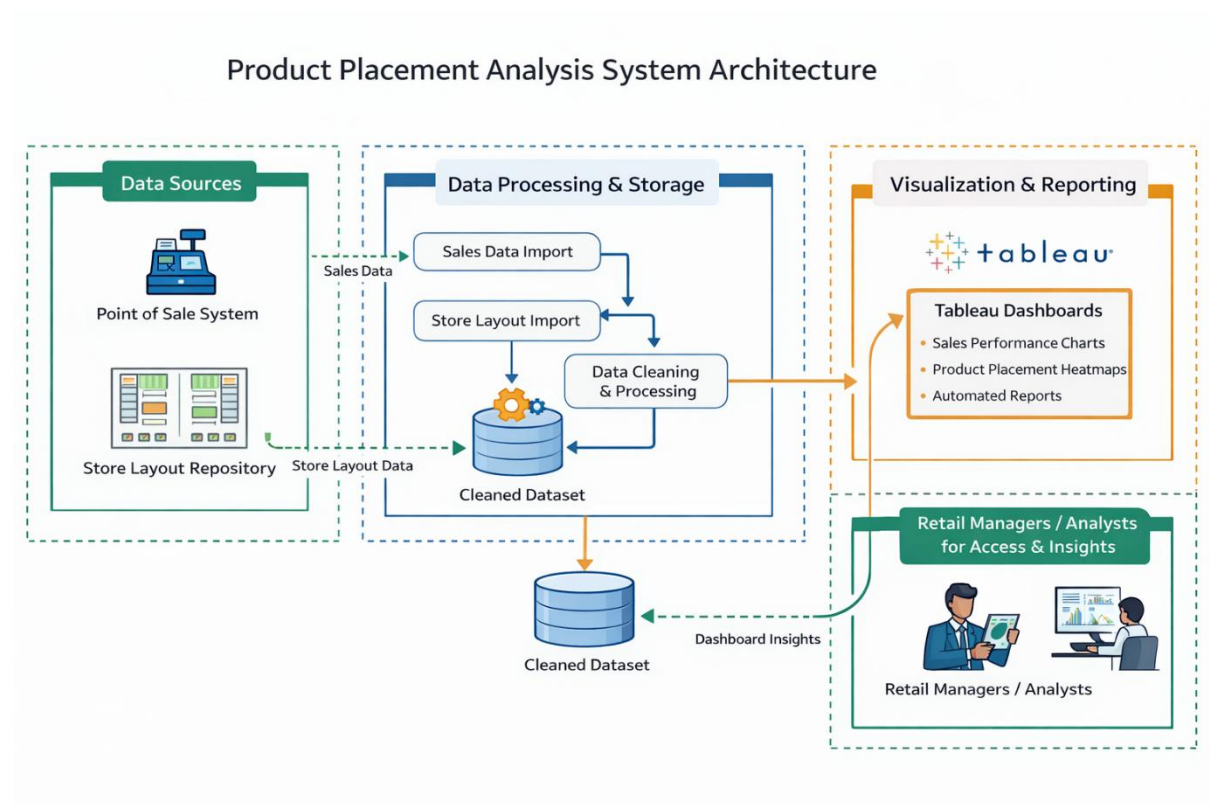


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinicalresearch-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>

5. PROJECT PLANNING & SCHEDULING :

5.1 Project Planning :

5.1.1 Planning Logic :

Project Title: Strategic Product Placement Analysis

Epic: Unveiling Sales Impact with Tableau Visualization

This epic is too large to complete in one sprint and will be broken down into smaller tasks over multiple sprints.

Sprint 1: Data Acquisition & Preparation (5 Days)

- Story: Data Collection
 - Collecting UNESCO World Heritage Site data: 2 Story Points (Very Easy task)
 - Loading data into a suitable format: 1 Story Point (Very Easy task)

- Story: Data Preprocessing
 - Handling missing values in the dataset: 3 Story Points (Moderate task)
 - Handling categorical values (e.g., criteria, types): 2 Story Points (Easy task)

Sprint 2: Tableau Development & Deployment (5 Days)

- Story: Dashboard Building
 - Designing and building interactive Tableau dashboards: 5 Story Points (Difficult task)
 - Testing and refining dashboard functionality: 3 Story Points (Moderate task)
- Story: Deployment & Presentation
 - Preparing Tableau workbook for sharing/embedding: 3 Story Points (Moderate task)
 - Documenting insights and findings for presentation: 5 Story Points (Difficult task)

Velocity Calculation:

- Total Story Points (Sprint 1): $2 + 1 + 3 + 2 = 8$ Story Points
- Total Story Points (Sprint 2): $5 + 3 + 3 + 5 = 16$ Story Points
- Total Story Points Completed: $8 + 16 = 24$ Story Points
- Number of Sprints: 2
- Velocity: $\text{Total Story Points Completed} / \text{Number of Sprints} = 24 / 2 = 12$ Story Points per Sprint

Your team's velocity is 12 Story Points per Sprint.

5.1.2 Project Planning :

The project planning for Strategic Product Placement Analysis begins with identifying business objectives and understanding the retailer's need to determine how product placement influences overall sales performance. The planning process includes collecting relevant datasets such as sales history, product details, and placement zone information. These datasets are examined for completeness and prepared through data cleaning, formatting, and standardization using tools like Python and Excel to ensure accuracy before analysis.

After preparing the data, the project progresses into the analytical and visualization phase. Here, patterns, correlations, and key performance indicators (KPIs) related to

placement effectiveness are defined. A visualization plan is created to outline the charts, heatmaps, KPIs, and comparisons required for the dashboard. Tableau is then used to build interactive dashboards that help users explore product placement impacts, category trends, and store-level variations through filters and intuitive layouts.

Finally, the planning includes testing outputs, validating dashboard results with the processed dataset, and refining visual designs based on feedback. Once complete, the dashboard is published through Tableau Public for easy access and sharing. The project concludes with compiling insights and actionable recommendations for retailers, ensuring the findings can directly support strategic decision-making and layout optimization.

6.FUNCTIONAL AND PERFORMANCE TESTING :

6.1 Performance Testing :

Functional testing for the project ensures that every component of the product placement analysis system works as intended. This includes validating whether the cleaned datasets load correctly into Tableau, checking that filters (such as category, product, store, or date) respond accurately, and verifying that each visualization—heatmaps, bar charts, KPIs, trend lines—displays the correct data. Functional testing also focuses on ensuring the user can interact with the dashboard smoothly, such as drilling down into specific categories or switching between comparison views without errors or broken visuals.

Performance testing focuses on the efficiency, speed, and stability of the dashboard under different conditions. This involves measuring how quickly Tableau renders visualizations when applying filters or loading large datasets. It also checks for delays or lag when switching between dashboard pages or interacting with multiple charts. The goal is to ensure that the dashboard loads quickly, handles large sales datasets efficiently, and maintains consistent responsiveness for end-users.

Together, functional and performance testing verify that the analytical system not only works correctly but also performs reliably under practical usage scenarios, ensuring smooth decision-making and a high-quality user experience for stakeholders.

7. RESULTS :

7.1 Output Screenshots :

Fig 1 : Avg. Sales Volume by Category

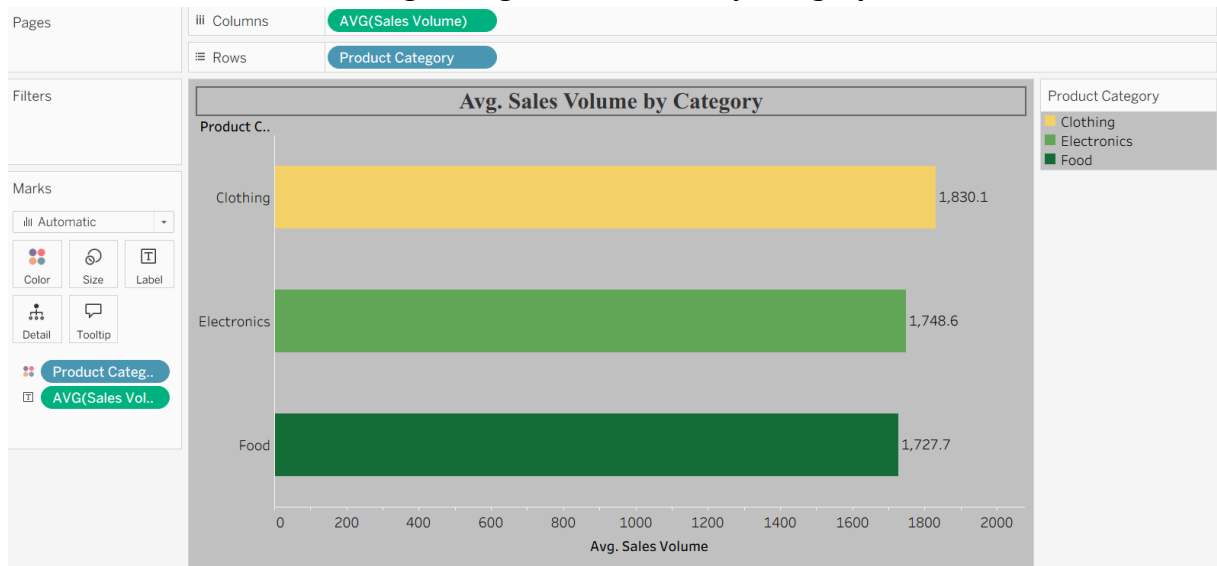


Fig 2 : Competitor Price VS Price



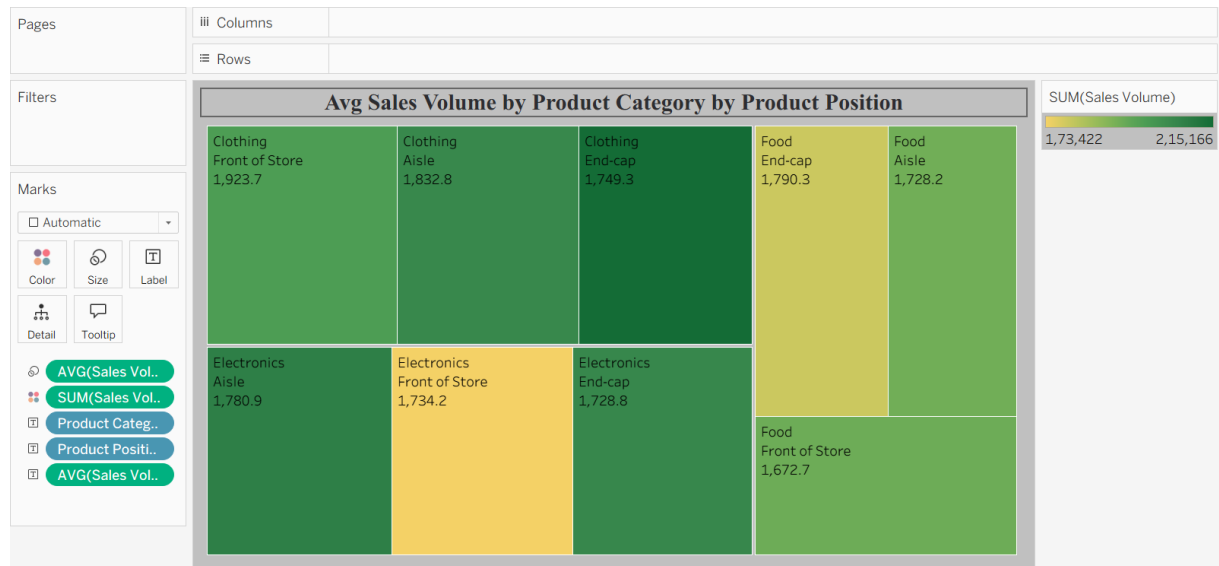
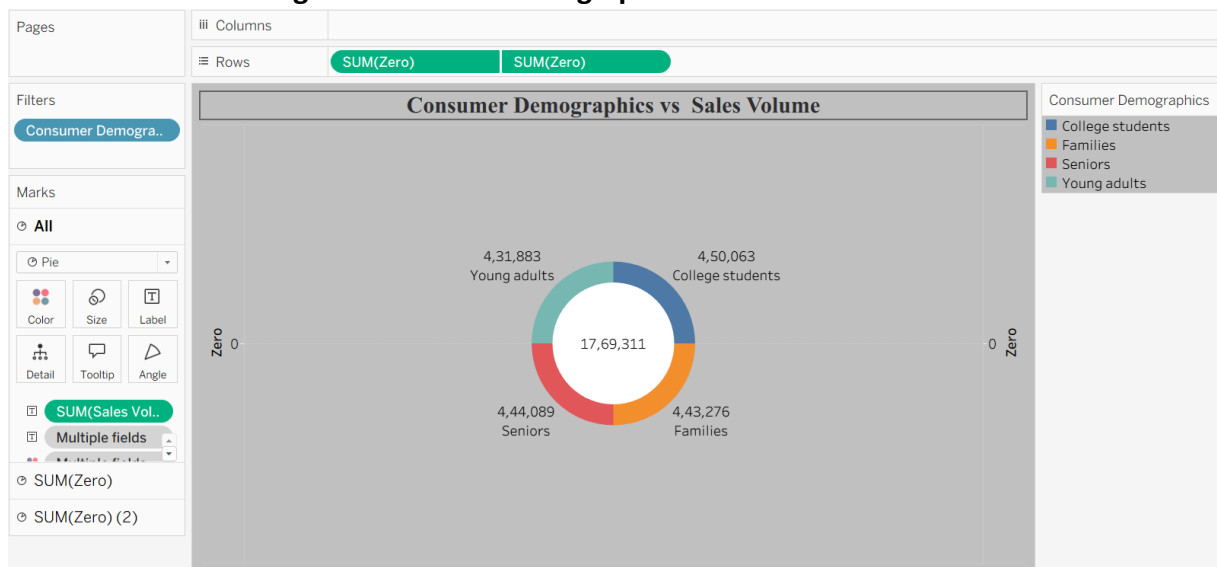
Fig 3 : Avg Sales Volume by Product Category by Product Position**Fig 4 : Consumer Demographics vs Sales Volume**

Fig 5 : Product Category vs Price

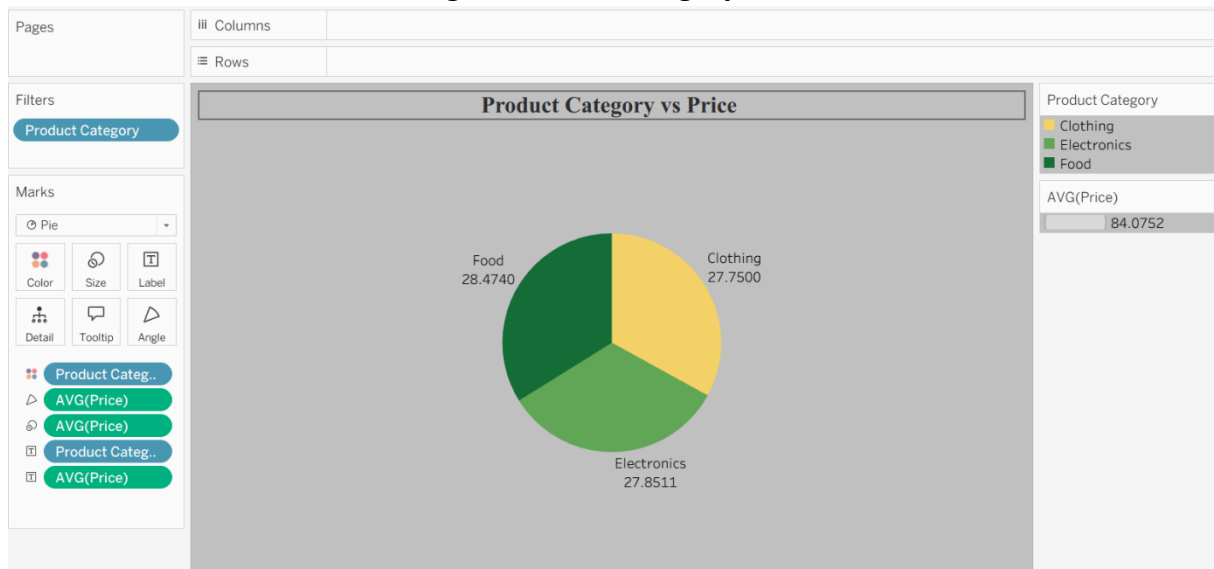


Fig 6 : Avg. Sales Volume by Product Category by Season

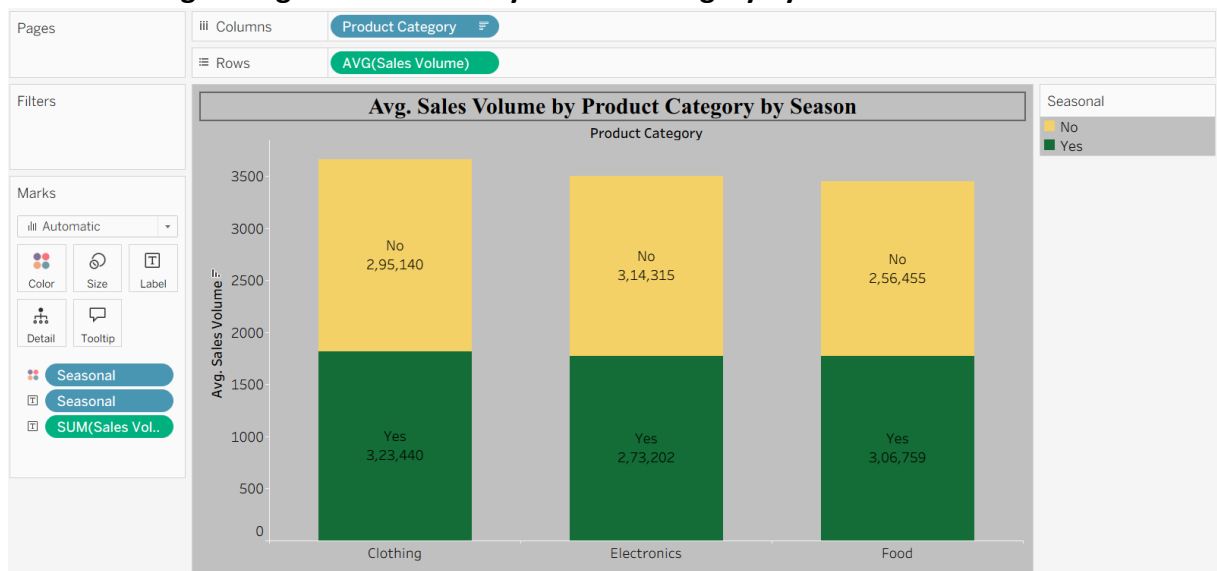


Fig 7 : Foot Traffic by Avg Sales Volume

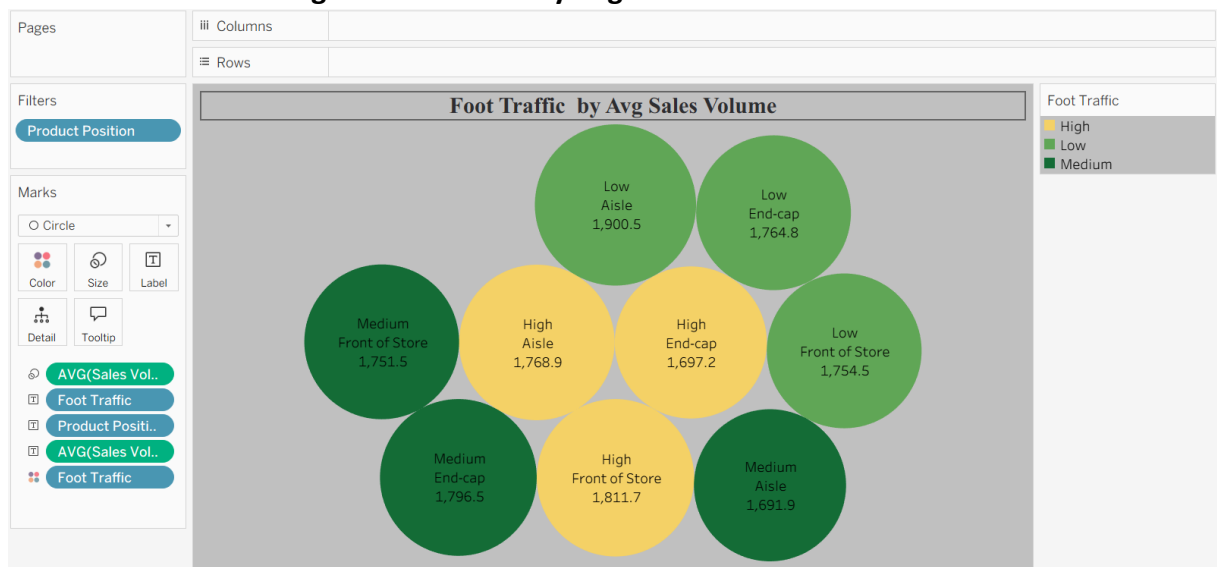
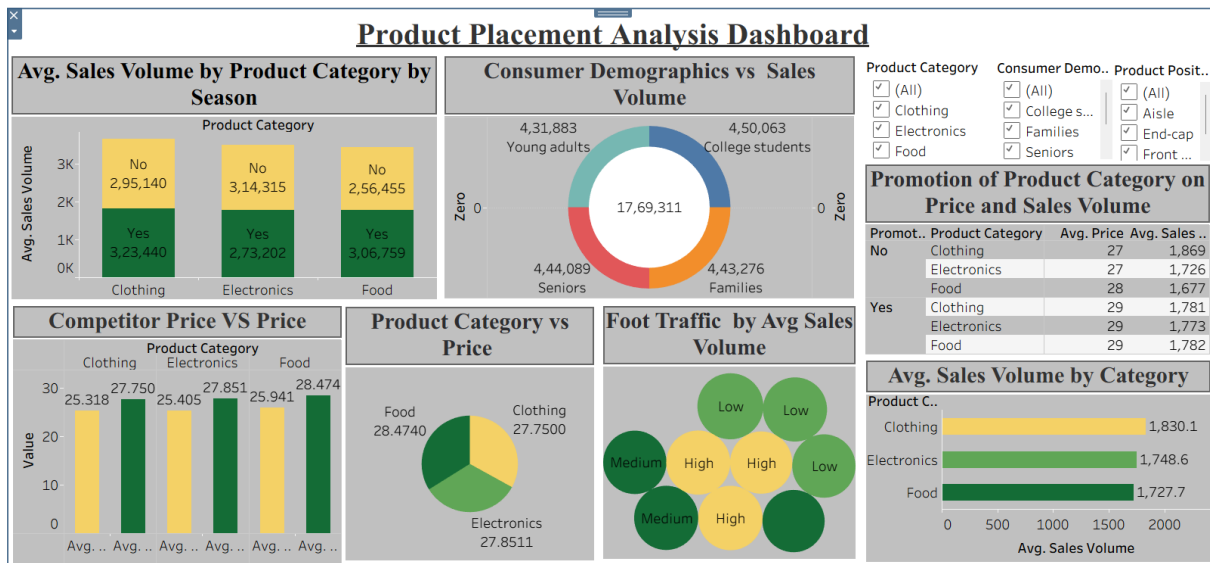


Fig 8 : Promotion of Product Category on Price and Sales Volume

Promotion		Product Category		Avg. Price	Avg. Sales Volume
No	Clothing			27	1,869
	Electronics			27	1,726
	Food			28	1,677
Yes	Clothing			29	1,781
	Electronics			29	1,773
	Food			29	1,782

7.2 Dashboard Screenshots :

Dashboard 1



8. Advantages and Disadvantages :

Advantages :

- Data-Driven Decision Making**
 Retailers can make accurate placement decisions based on real sales patterns rather than assumptions.
- Improved Sales Performance**
 Identifies high-impact zones/shelves that maximize product visibility and increase revenue.
- Interactive Visual Insights**
 Tableau dashboards provide user-friendly, real-time filtering and interactive analysis.
- Quick Identification of Trends**
 Helps detect seasonal trends, category performance, and placement-based behavior instantly.
- Enhanced Store Layout Optimization**
 Supports optimizing shelf arrangements and store layouts for better customer engagement.
- Better Resource Allocation**
 Helps retailers focus on high-performing products and zones, saving time and effort.

- **Easy Sharing & Accessibility**

Dashboards published via Tableau Public can be accessed anywhere without complex software installation.

Disadvantages :

- **Dependent on Data Quality**

Poor or incomplete sales/placement data can lead to inaccurate insights.

- **No Predictive Modeling**

Tableau visualizes current trends but cannot predict future placement performance without additional ML models.

- **Limited Custom Logic in Tableau**

Complex statistical operations cannot be directly implemented within Tableau alone.

- **Requires Skilled Users**

Non-technical users may struggle with interacting with filters and advanced dashboard features.

- **Performance Issues With Large Data**

Very large datasets may cause slow dashboard loading or lag in interactions.

- **Static Placement Categories**

If placement zones change frequently, visualizations require regular updates.

- **Not Fully Automated**

Data preprocessing, cleaning, and updates often require manual intervention.

9.Conclusion :

The project *Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization* successfully demonstrates how data-driven insights can significantly enhance retail decision-making. By integrating sales data, product information, and placement details, the system provides a clear understanding of how product positioning influences customer purchasing behavior. Through structured data preprocessing and analytical planning, the project transforms raw data into meaningful and actionable insights.

The interactive Tableau dashboard developed in the project plays a crucial role in visualizing placement effectiveness, category-wise performance, and sales trends. Its dynamic filters, heatmaps, KPIs, and comparison charts enable users to explore data intuitively and uncover hidden patterns that traditional reporting methods often overlook. This real-time visualization capability empowers store managers, analysts, and decision-makers to identify high-performing zones and optimize store layouts efficiently.

Overall, the project highlights the value of combining structured analytics with advanced visualization tools to support strategic retail planning. The insights generated not only help in improving product placement strategies but also contribute to enhanced customer engagement and increased revenue potential. With further enhancements such as predictive

modeling and automated data updates, this system can evolve into a comprehensive decision-support platform for modern retail environments.

10.Future Scope :

The project has strong potential for expansion through advanced analytics and automation. One major future enhancement is the integration of **predictive modeling** and **machine learning algorithms** to forecast sales performance based on different product placement scenarios. This would allow retailers to simulate layout changes before implementation and choose the best arrangement for maximum sales impact.

Another important future direction is the development of a **real-time data pipeline**. Automating data collection from POS systems and integrating it directly with Tableau dashboards would ensure that insights are always up-to-date without manual intervention. This enables continuous monitoring of product placement effectiveness and faster response to market changes.

Additionally, the system can be extended to include **customer behavior analytics**, such as footfall heatmaps or in-store navigation patterns, by integrating IoT sensors or camera analytics. Expanding the dashboards to support **multi-store comparison**, **automated report generation**, and **mobile-friendly visualization** will further improve usability and make the tool a robust decision-support system for large-scale retail operations.

11.Appendix :

Dataset Link :-

- <https://www.kaggle.com/datasets/amitvkulkarni/impact-of-product-positioning-on-sales>

Git Hub Link :-

- <https://github.com/Pavankalyan-12395/Strategic-Product-Placement-Analysis.git>

Tableau File Link :-

- https://public.tableau.com/views/ProductPlacementAnalysisProject/Story1?:language=enUS&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Web Integration Video Link :

- <https://drive.google.com/file/d/1yR63ZAW2SMhkt9T5diqr6Mu6QV5TSIFe/view?usp=sharing>